

THEORETICAL MODEL: $Sales = \beta_0 + \beta_1 Price + \beta_2 Price.Comp + \varepsilon$

ESTIMATED MODEL: $\hat{Sales}_i = 119.3 - 1.28 Price_i + 1.04 Price.Comp_i$

most element identical
to simple linear regression

Call:

```
lm(formula = Sales ~ Price + Price.Comp, data = Sales)
```

Residuals:

Min	1Q	Median	3Q	Max
-46.131	-12.662	-1.136	11.909	51.358

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	119.31366	10.38034	11.49	< 2e-16 ***
Price	-1.28361	0.13085	-9.81	3.45e-16 ***
Price.Comp	1.03933	0.08958	11.60	< 2e-16 ***

significance
of
coefficients

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.26 on 97 degrees of freedom

Multiple R-squared: 0.739, Adjusted R-squared: 0.7336

F-statistic: 137.3 on 2 and 97 DF, p-value: < 2.2e-16

$$s_e = \sqrt{\frac{SSE}{n-k-1}}$$

$$n-k-1$$

$$Adjusted-R^2 = 1 - \frac{SSE / (n-k-1)}{SST / (n-1)}$$



- modification of R^2 that penalises overparametrisation
- use this to compare models with different number of parameters (k)

Test for overall significance.

p-value < $2.2 \cdot 10^{-16}$ for the test:

$$H_0: \beta_1 = \beta_2 = 0$$

$$H_1: \text{At least one coefficient} \neq 0$$

We reject H_0 . The model

is overall significant

Note that the p-value is

calculated from the F-score = 137.3

which under H_0 comes from

an $F_{2,97}$ distribution